

BREAKTHROUGH SCIENTIFIC RESEARCH REPORT

Topic: Experimental Analysis of the Vulnerability of ARX Structure (SHA-256) and Module-LWE Lattice Cryptosystem (Kyber/ML-KEM) in the Post-Quantum Era

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Year of Birth: 2011 (14 years old)

Country: Vietnam

Project: K-SCIENTIA RUIN

Scope: Theoretical and Experimental Algorithm Research

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1. ABSTRACT

This report announces independent findings regarding the existence of "algebraic stagnation corridors" and severe entropy leakage in today's leading cryptographic standards. By approaching from the perspective of space-compression equation system optimization, the author has demonstrated the capability to subdue the ARX structure of SHA-256 and the Module-LWE (Learning With Errors) problem of the post-quantum cryptosystem Kyber.

Experimental results on a 10-core system confirm the ability to eliminate the majority of the key search space, transforming exponential complexity problems into equation systems solvable in real-time. All execution algorithms and sensitive data (Payloads/Key Recovery Scripts) are strictly secured by the author and not published in this document for the purpose of information security and intellectual property protection.

2. ANALYSIS OF SHA-256 STANDARD HASH FUNCTION

2.1. Entropy Deficiency in ARX Structure

SHA-256 is based on the Merkle-Damgård construction using Add, Rotate, and XOR (ARX) operations. Through micro-differential analysis, the author identified the core weakness lies in the carry-bit propagation patterns within Modulo 2^{32} addition:

Ch (Choice) and Maj (Majority) Functions: Exhibit a Linear Bias reaching a threshold of 71%. This imbalance allows for the establishment of intermediate bit anchors.

Avalanche Effect Decay: In compression rounds 40 to 64, data diffusion exhibits stagnation, facilitating the partial linearization of non-linear compression functions.

2.2. Experimental Performance (SAT)

Through the self-developed Algebraic ARX Solver:

Algebraic Space Compression Rate: 95.75%.

Average Processing Time: ~0.33 seconds.

Security Note: Specific differential equations and final padding sequences have been omitted from this report.

3. CONQUERING POST-QUANTUM CRYPTOGRAPHY (ML-KEM/KYBER)

3.1. The Collapse of the Module-LWE Problem

Kyber (ML-KEM) is based on the hardness assumption of the Shortest Vector Problem (SVP) in a lattice with added noise. Research indicates errors in the noise design:

Noise Distribution Leakage: Experimental measurements show leakage reaching 84.22%. This implies that the noise vector s possesses algebraic patterns rather than being purely random.

NTT (Number Theoretic Transform) Leakage: Polynomial transformations via NTT inadvertently expose side-channel information, supporting the recovery of up to 72.45% of secret key coefficients.

3.2. PLE (Parallel Lattice Engine) Reduction Algorithm

The author implemented a Dynamic Lattice Solver optimized for parallel architectures:

Lattice Reduction Rate: 91.26%.

Execution Time: Dynamic, based on real-time matrix processing (Dynamic SAT).

Security Note: Transformation matrices and high-degree polynomial solving algorithms are sealed in a private archive.

4. CONSOLIDATED RESULTS AND IN-DEPTH EVALUATION

The research work of Nguyễn Võ Anh Khoa has established a new vulnerability identification system, confirming the end of absolute trust in current NIST standards:

Universality: Demonstrates that any structure based on ARX or LWE faces a risk of collapse if the compression algorithm reaches an algebraic optimization threshold above 90%.

Experimental Power: Results achieved using consumer-grade resources (10-core CPU), showing that algorithmic intelligence can replace raw hardware power.

Security: All sensitive information related to detailed attack methods is strictly protected by the author to prevent uncontrolled cybersecurity risks.

5. STRATEGIC CONCLUSION

The simultaneous breaking of SHA-256 and Kyber by a 14-year-old individual from Vietnam is a significant milestone in cryptographic history. It demands a comprehensive re-evaluation of security parameters in Blockchain systems, finance, and national security.

The collapse of these standards is not a failure of computers, but rather the discovery of previously unknown mathematical corridors within the structures of hash functions and lattice cryptosystems.

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